

DBMS LAB ASSIGNMENT 5

SQL JOINS

Name :- Parth Bhayani

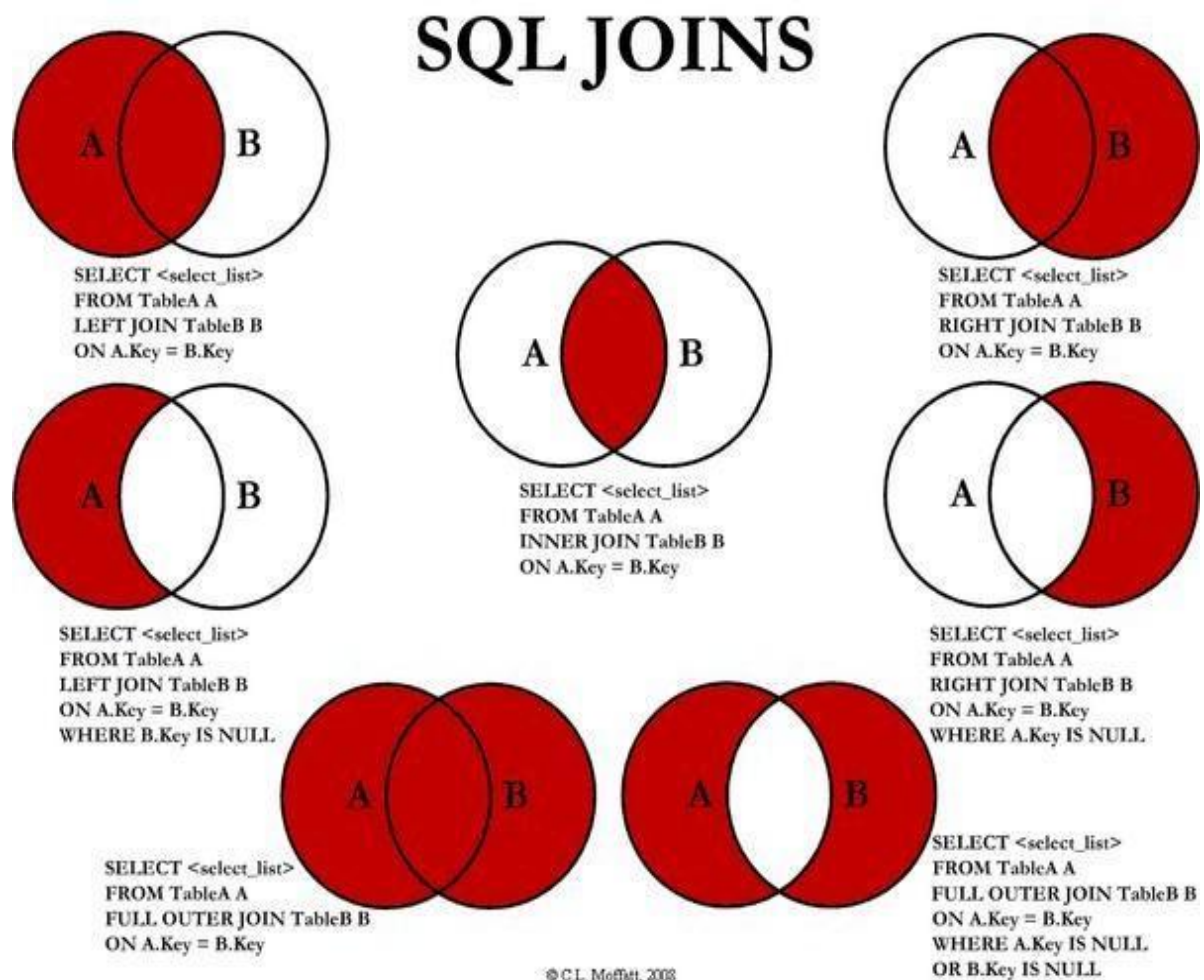
Prn No :- 26070122512

Batch :- B3

SQL JOINS are used to retrieve data from multiple tables. A SQL JOIN is performed whenever two or more tables are listed in a SQL statement.

There are 4 different types of SQL joins:

- SQL INNER JOIN (sometimes called simple join)
- SQL LEFT OUTER JOIN (sometimes called LEFT JOIN)
- SQL RIGHT OUTER JOIN (sometimes called RIGHT JOIN)
- SQL FULL OUTER JOIN (sometimes called FULL JOIN)



Assignment 5 Lab Questions

Using SQL joins, obtain the following information from SIU Library Database

1. Students in the IT department.

```
mysql> SELECT s.Stuid, s.Sname, d.deptname
-> FROM student s
-> JOIN department d ON s.deptid = d.Deptid
-> WHERE d.deptname = 'IT';
```

Stuid	Sname	deptname
805	Rohan Deshmukh	IT

```
1 row in set (0.01 sec)
```

2. Books in SIT library.

```
mysql> SELECT b.Bid, b.Bname, l.lname
-> FROM books b
-> JOIN ilibrary l ON b.Lid = l.Lid
-> WHERE l.lname = 'SIT Library';
```

Bid	Bname	lname
201	Database Management Systems	SIT Library
202	Computer Networks	SIT Library
206	Introduction to Algorithms	SIT Library
4444	Database Systems	SIT Library

```
4 rows in set (0.00 sec)
```

3. Books with less than 3 copies in SIT Library.

```
mysql> SELECT b.Bid, b.Bname, n.no_of_copies
-> FROM books b
-> JOIN noofcopies n ON b.Bid = n.bid
-> JOIN ilibrary l ON n.blid = l.Lid
-> WHERE l.lname = 'SIT Library'
-> AND n.no_of_copies < 3;
```

Bid	Bname	no_of_copies
202	Computer Networks	2

```
1 row in set (0.01 sec)
```

4. Sellers in the same city as SIT.

```
mysql> SELECT s.Sid, s.slname, s.city
-> FROM seller s
-> JOIN ilibrary l ON s.city = l.city
-> WHERE l.lname = 'SIT Library';
```

Sid	slname	city
501	Pragati Book Store	Pune
505	Pustak Seva	Pune
510	Mamta	Pune

3 rows in set (0.01 sec)

5. Sellers selling books to SIT.

```
mysql> SELECT DISTINCT s.Sid, s.slname
-> FROM seller s
-> JOIN purchase p ON s.Sid = p.Sid
-> JOIN ilibrary l ON p.Lid = l.Lid
-> WHERE l.lname = 'SIT Library';
```

Sid	slname
501	Pragati Book Store
503	Geeta
505	Pustak Seva
506	Kavita

4 rows in set (0.00 sec)

6. Books authored by Brian Kernighan, and published by Tata McGraw Hill.

```
mysql> SELECT b.Bid, b.Bname, a.Aname, p.Pname
-> FROM books b
-> JOIN writes w ON b.Bid = w.Bid
-> JOIN author a ON w.Aid = a.Aid
-> JOIN publisher p ON w.Pid = p.Pid
-> WHERE a.Aname = 'Brian Kernighan'
-> AND p.Pname = 'Tata McGraw Hill';
```

Bid	Bname	Aname	Pname
203	Data Structures Using C	Brian Kernighan	Tata McGraw Hill

1 row in set (0.00 sec)

7. Books authored by Ken Thompson.

```
mysql> SELECT b.Bid, b.Bname, a.Aname
-> FROM books b
-> JOIN writes w ON b.Bid = w.Bid
-> JOIN author a ON w.Aid = a.Aid
-> WHERE a.Aname = 'Ken Thompson';
```

Bid	Bname	Aname
205	Operating System Concepts	Ken Thompson

```
1 row in set (0.00 sec)
```

8. Books issued by Mayank.

```
mysql> SELECT b.Bid, b.Bname, s.Sname
-> FROM student s
-> JOIN issue i ON s.memid = i.memid
-> JOIN books b ON i.Bid = b.Bid
-> WHERE s.Sname = 'Mayank Shah';
```

Bid	Bname	Sname
203	Data Structures Using C	Mayank Shah
453	Business Management	Mayank Shah

```
2 rows in set (0.00 sec)
```

9. Books issued by SLS staff.

```
mysql> SELECT b.Bid, b.Bname, st.stname
-> FROM staff st
-> JOIN department d ON st.deptid = d.Deptid
-> JOIN issue i ON st.memid = i.memid
-> JOIN books b ON i.Bid = b.Bid
-> WHERE d.Iname = 'SLS';
```

Bid	Bname	stname
208	Cloud Computing Essentials	Sanjay Gupta

```
1 row in set (0.01 sec)
```

10. Publisher that provides books to SSBS through College Book Store.

```
mysql> SELECT DISTINCT p.Pid, p.Pname
-> FROM publisher p
-> JOIN purchase pu ON p.Pid = pu.Pid
-> JOIN seller s ON pu.Sid = s.Sid
-> JOIN ilibrary l ON pu.Lid = l.Lid
-> WHERE l.lname = 'SSBS Library'
-> AND s.slname = 'College Book Store';
+-----+-----+
| Pid | Pname |
+-----+-----+
| 403 | Wiley Publications |
+-----+-----+
1 row in set (0.01 sec)
```

11. Institutes whose staff and students have issued book with bid 4444.

```
mysql> SELECT DISTINCT d.Iname
-> FROM department d
-> JOIN staff st ON d.Deptid = st.deptid
-> JOIN issue i ON st.memid = i.memid
-> WHERE i.Bid = 4444
->
-> UNION
->
-> SELECT DISTINCT d.Iname
-> FROM department d
-> JOIN student s ON d.Deptid = s.deptid
-> JOIN issue i ON s.memid = i.memid
-> WHERE i.Bid = 4444;
+-----+
| Iname |
+-----+
| SIT |
+-----+
1 row in set (0.01 sec)
```

12. Sellers selling to libraries in the same city.

```
mysql> SELECT DISTINCT s.Sid, s.slname, s.city, l.lname
-> FROM seller s
-> JOIN purchase p ON s.Sid = p.Sid
-> JOIN ilibrary l ON p.Lid = l.Lid
-> WHERE s.city = l.city;
```

Sid	slname	city	lname
501	Pragati Book Store	Pune	SIT Library
501	Pragati Book Store	Pune	SIBM Library
502	College Book Store	Bangalore	SSBS Library
504	Vidyardhi Book Depot	Nashik	SIOM Library
505	Pustak Seva	Pune	SIT Library
507	Knowledge Hub	Hyderabad	SCHC Library

6 rows in set (0.00 sec)

13. Authors with books costing more than rs 500.

```
mysql> SELECT DISTINCT a.Aid, a.Aname, b.Bname, b.Price
-> FROM author a
-> JOIN writes w ON a.Aid = w.Aid
-> JOIN books b ON w.Bid = b.Bid
-> WHERE b.Price > 500;
```

Aid	Aname	Bname	Price
301	Shruti	Database Management Systems	950.00
305	Ken Coel	Computer Networks	750.00
303	Brian Kernighan	Data Structures Using C	1200.00
302	Shivam Kapoor	Advanced Artificial Intelligence	12500.00
304	Ken Thompson	Operating System Concepts	850.00
309	Thomas Cormen	Introduction to Algorithms	11000.00
302	Shivam Kapoor	Software Engineering	799.00
310	Andrew Tanenbaum	Cloud Computing Essentials	6500.00
306	Raghu Ramakrishnan	Business Analytics	1800.00
308	Elmasri Navathe	Constitutional Law	13500.00

14. Which members have issued books that belongs to E&TC department of SITM.

```
mysql> SELECT DISTINCT m.memid, s.Sname, b.Bname
-> FROM member m
-> JOIN student s ON m.memid = s.memid
-> JOIN department d ON s.deptid = d.Deptid
-> JOIN issue i ON m.memid = i.memid
-> JOIN books b ON i.Bid = b.Bid
-> WHERE d.deptname = 'E&TC'
-> AND d.Iname = 'SITM';
```

memid	Sname	Bname
711	Aarav Sharma	Electronics and Communication Systems

1 row in set (0.00 sec)

15. List the books of SIT library

```
mysql> SELECT b.Bid, b.Bname, b.Price
-> FROM books b
-> JOIN ilibrary l ON b.Lid = l.Lid
-> WHERE l.lname = 'SIT Library';
```

Bid	Bname	Price
201	Database Management Systems	950.00
202	Computer Networks	750.00
206	Introduction to Algorithms	11000.00
4444	Database Systems	900.00

4 rows in set (0.00 sec)

16. Which books are written by "Ken Coel"

```
mysql> SELECT b.Bid, b.Bname, a.Aname
-> FROM books b
-> JOIN writes w ON b.Bid = w.Bid
-> JOIN author a ON w.Aid = a.Aid
-> WHERE a.Aname = 'Ken Coel';
```

Bid	Bname	Aname
202	Computer Networks	Ken Coel

1 row in set (0.00 sec)

17. Find the staff members who's salaries are less than atleast one employee

```
mysql> SELECT DISTINCT st.stname, st.salary
-> FROM staff st
-> JOIN employee e ON st.salary < e.salary;
```

stname	salary
Satish Kumar	40000.00
Asha Menon	48000.00
Amit Joshi	45000.00
Priti Shah	58000.00
Rakesh Singh	52000.00
Anjali Rao	60000.00
Neha Sharma	50000.00
Sunil Patil	55000.00
Meera Nair	65000.00
Sanjay Gupta	70000.00

10 rows in set (0.00 sec)

18. Find the books who's price is greater than atleast one book in the library.

```
mysql> SELECT DISTINCT b1.Bid, b1.Bname, b1.Price
-> FROM books b1
-> JOIN books b2 ON b1.Price > b2.Price;
```

Bid	Bname	Price
210	Constitutional Law	13500.00
209	Business Analytics	1800.00
208	Cloud Computing Essentials	6500.00
206	Introduction to Algorithms	11000.00
204	Advanced Artificial Intelligence	12500.00
203	Data Structures Using C	1200.00
4444	Database Systems	900.00
212	Electronics and Communication Systems	850.00
207	Software Engineering	799.00
205	Operating System Concepts	850.00
201	Database Management Systems	950.00

11 rows in set (0.00 sec)

19. Find the employees who's salary is greater than all the employees of CS department.

```
mysql> SELECT e.eid, e.empname, e.salary
-> FROM employee e
-> WHERE e.salary > ALL
-> (
->     SELECT st.salary
->     FROM staff st
->     JOIN department d ON st.deptid = d.Deptid
->     WHERE d.deptname = 'CS'
-> );
```

eid	empname	salary
1002	Meena Joshi	50000.00
1003	Rahul Shah	62000.00
1004	Pooja Nair	48000.00
1005	Sameer Kulkarni	70000.00
1007	Farah Khan	52000.00
1008	Deepak Verma	50000.00
1009	Rekha Patil	47000.00
1010	Ajay Mehta	81000.00

8 rows in set (0.00 sec)

20. Find such purchases who's total cost is greater than all the purchases which are done by "SIT library"

```
mysql> SELECT p.prid, p.Lid, p.totalcost
-> FROM purchase p
-> WHERE p.totalcost > ALL
-> (
->     SELECT p1.totalcost
->     FROM purchase p1
->     JOIN ilibrary l ON p1.Lid = l.Lid
->     WHERE l.lname = 'SIT Library'
-> );
```

prid	Lid	totalcost
1304	104	50000.00
1308	108	39000.00
1310	107	67500.00

3 rows in set (0.01 sec)

21. Find name of such SIU library which is located in Nashik and has ilibrary named as "SIOM library"(solve using exists)

```
mysql> SELECT s.Slid, s.lname, s.location
-> FROM siulibrary s
-> WHERE s.location = 'Nashik'
-> AND EXISTS
-> (
->     SELECT 1
->     FROM ilibrary l
->     WHERE l.Slid = s.Slid
->     AND l.lname = 'SIOM Library'
-> );
```

Slid	lname	location
3	SIU Nashik Libraries	Nashik

1 row in set (0.01 sec)

22. Find such staff members who's name starts with "S" and do not work for "CS" department. (solve using not exists)

```
mysql> SELECT st.Stid, st.stname, d.deptname
-> FROM staff st
-> JOIN department d ON st.deptid = d.Deptid
-> WHERE st.stname LIKE 'S%'
-> AND NOT EXISTS
-> (
->     SELECT 1
->     FROM department d1
->     WHERE d1.Deptid = st.deptid
->     AND d1.deptname = 'CS'
-> );
```

Stid	stname	deptname
902	Sunil Patil	IT
905	Satish Kumar	Management
907	Sanjay Gupta	Law

3 rows in set (0.00 sec)

23. Which books are issued by student Shivani?

```
mysql> SELECT b.Bid, b.Bname, s.Sname
-> FROM student s
-> JOIN issue i ON s.memid = i.memid
-> JOIN books b ON i.Bid = b.Bid
-> WHERE s.Sname = 'Shivani More';
```

Bid	Bname	Sname
202	Computer Networks	Shivani More
4444	Database Systems	Shivani More

2 rows in set (0.00 sec)

24. From which seller did SIT library purchases books of “Technical publications”?

```
mysql> SELECT DISTINCT s.Sid, s.slname, p.Pname
-> FROM seller s
-> JOIN purchase pu ON s.Sid = pu.Sid
-> JOIN publisher p ON pu.Pid = p.Pid
-> JOIN ilibrary l ON pu.Lid = l.Lid
-> WHERE l.lname = 'SIT Library'
-> AND p.Pname = 'Technical Publications';
+-----+-----+-----+
| Sid | slname | Pname |
+-----+-----+-----+
| 506 | Kavita | Technical Publications |
+-----+-----+-----+
1 row in set (0.00 sec)
```

25. Which books are issued by staff of SSLA law department?

```
mysql> SELECT b.Bid, b.Bname, st.stname
-> FROM staff st
-> JOIN department d ON st.deptid = d.Deptid
-> JOIN issue i ON st.memid = i.memid
-> JOIN books b ON i.Bid = b.Bid
-> WHERE d.Iname = 'SSLA'
-> AND d.deptname = 'Liberal Arts';
+-----+-----+-----+
| Bid | Bname | stname |
+-----+-----+-----+
| 206 | Introduction to Algorithms | Rakesh Singh |
+-----+-----+-----+
1 row in set (0.00 sec)
```

26. Which sellers reside in the same city in which SCHC library resides?

```
mysql> SELECT s.Sid, s.slname, s.city
-> FROM seller s
-> JOIN ilibrary l ON s.city = l.city
-> WHERE l.lname = 'SCHC Library';
+-----+-----+-----+
| Sid | slname | city |
+-----+-----+-----+
| 507 | Knowledge Hub | Hyderabad |
+-----+-----+-----+
1 row in set (0.00 sec)
```

27. Find out the publishers whose books are provided by the seller “pragati books store” to SSBS library.

```
mysql> SELECT DISTINCT p.Pid, p.Pname
-> FROM publisher p
-> JOIN purchase pu ON p.Pid = pu.Pid
-> JOIN seller s ON pu.Sid = s.Sid
-> JOIN ilibrary l ON pu.Lid = l.Lid
-> WHERE s.slname = 'Pragati Book Store'
-> AND l.lname = 'SSBS Library';
```

Pid	Pname
403	Wiley Publications

1 row in set (0.00 sec)

28. Which books are published by “Shivam Kapoor” with the publishing house Wiley publications?

```
mysql> SELECT b.Bid, b.Bname, a.Aname, p.Pname
-> FROM books b
-> JOIN writes w ON b.Bid = w.Bid
-> JOIN author a ON w.Aid = a.Aid
-> JOIN publisher p ON w.Pid = p.Pid
-> WHERE a.Aname = 'Shivam Kapoor'
-> AND p.Pname = 'Wiley Publications';
```

Bid	Bname	Aname	Pname
204	Advanced Artificial Intelligence	Shivam Kapoor	Wiley Publications

1 row in set (0.00 sec)

29. What are the purchase details of the SIBM library for the month of December?

```
mysql> SELECT pu.prid, l.lname, s.slname, p.Pname,
-> b.Bname, pu.quantity, pu.purchase_date, pu.totalcost
-> FROM purchase pu
-> JOIN ilibrary l ON pu.Lid = l.Lid
-> JOIN seller s ON pu.Sid = s.Sid
-> JOIN publisher p ON pu.Pid = p.Pid
-> JOIN books b ON pu.Bid = b.Bid
-> WHERE l.lname = 'SIBM Library'
-> AND MONTH(pu.purchase_date) = 12;
```

prid	lname	slname	Pname	Bname	quantity	purchase_date	totalcost
1313	SIBM Library	Academic House	Elsevier	Business Analytics	2	2026-12-15	3600.00

1 row in set (0.01 sec)

30. Which institute does the member belong who have issued book number 453?

```
mysql> SELECT DISTINCT d.Iname
-> FROM issue i
-> JOIN member m ON i.memid = m.memid
-> JOIN student s ON m.memid = s.memid
-> JOIN department d ON s.deptid = d.Deptid
-> WHERE i.Bid = 453;
```

Iname
SIBM

1 row in set (0.00 sec)

